BIOL 3400 — Diagnostic Test Simulation

Beans, Coins & Dice: Building a 2×2 Table by Hand

Materials

- 1 bag with 80 beans: 16 red (sick) and 64 black (healthy)
- 1 coin (for testing sick patients)
- 1 standard six-sided die (for testing healthy patients)
- This worksheet and a pen

Setup

Prevalence: 20% of the population has the disease

- **Red bean** = Disease present (sick)
- **Black bean** = Disease absent (healthy)

The test:

- **If you draw a red bean (sick):** flip the coin. Heads = Test positive, Tails = Test negative.
  Sensitivity = 50%
- **If you draw a black bean (healthy):** roll the die. Roll a 1 = Test positive, anything else = Test negative.
  False positive rate ≈ 17%

Procedure (20 Draws, With Replacement)

1. Draw one bean from the bag without looking. Record the color.
2. If RED: flip the coin. Record Heads (Test +) or Tails (Test −).
3. If BLACK: roll the die. Record 1 (Test +) or 2–6 (Test −).
4. Put the bean back in the bag and mix well.
5. Repeat until you have 20 draws.

Recording Sheet

Draw #	Bean Color (Red / Black)	Disease Status (+ / −)	Test Procedure	Test Result (+ / −)
1				
2				
3				
4				
5				
6				
7				
8				
9				

10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

Step 1: Build Your 2×2 Table

Count your results from the recording sheet and fill in the table below:

	Disease +	**Disease −**	**Total**
Test +			
Test −			
Total			

Step 2: Compute the Metrics

Sensitivity = TP / (TP + FN) = _____ / _____ = _____
(Of the truly sick patients, what fraction did the test catch?)

Specificity = TN / (TN + FP) = _____ / _____ = _____
(Of the truly healthy patients, what fraction did the test correctly clear?)

False Discovery Rate (FDR) = FP / (TP + FP) = _____ / _____ = _____
(Of all positive tests, what fraction are wrong?)

Positive Predictive Value (PPV) = TP / (TP + FP) = _____ / _____ = _____
(If you test positive, what is the probability you are actually sick?)

Part 3: Discussion

Write your group's TP, FP, FN, TN on the board.

Q1: Is the FDR higher or lower than you expected? Why?

Q2: Looking at all groups' results on the board — are the numbers identical? Why not?

Q3: Would this be a good test to screen an entire population? Why or why not?

Step 4: Applying Bayes' Theorem

Now let's derive the answer your table already gave you, using Bayes' theorem.

What do we know from our experiment?

P(D) = _____ P(~D) = _____

P(Test+ | D) = _____ P(Test+ | ~D) = _____

Step 1: What is the total probability of testing positive?

P(Test+) = P(Test+ | D) × P(D) + P(Test+ | ~D) × P(~D)

= _____ × _____ + _____ × _____

= _____

Step 2: Apply Bayes' theorem.

P(D | Test+) = P(Test+ | D) × P(D) / P(Test+)

= _____ / _____

= _____

Step 3: Compare to your table.

Bayes answer: _____ Your empirical PPV: _____ Close? Y / N

Q4: The HIV test has 99% sensitivity, 95% specificity, but prevalence is only 0.3%. Predict: will the PPV be higher or lower than your experiment?